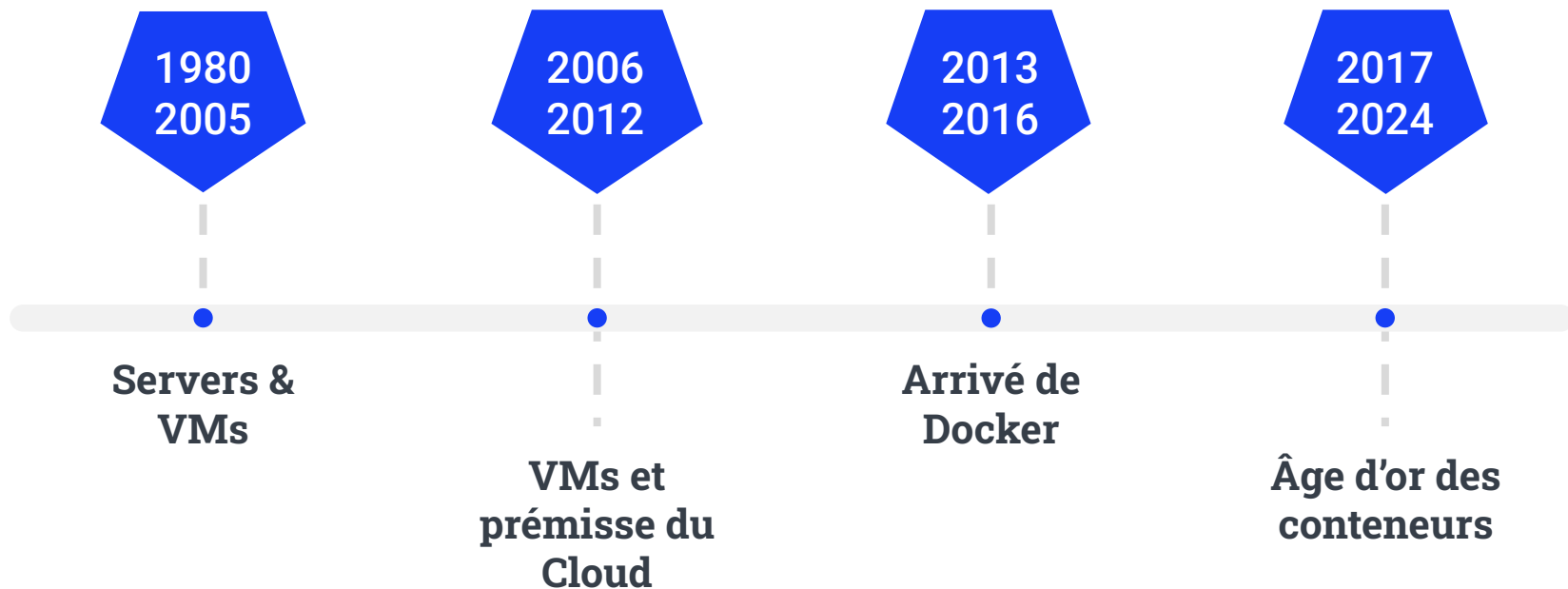


 **La Plateforme**

La grande école du numérique pour tous

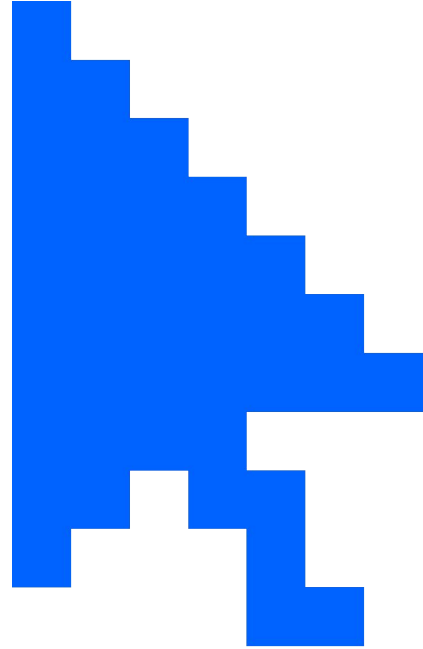
VM & DOCKER





Virtual Machine

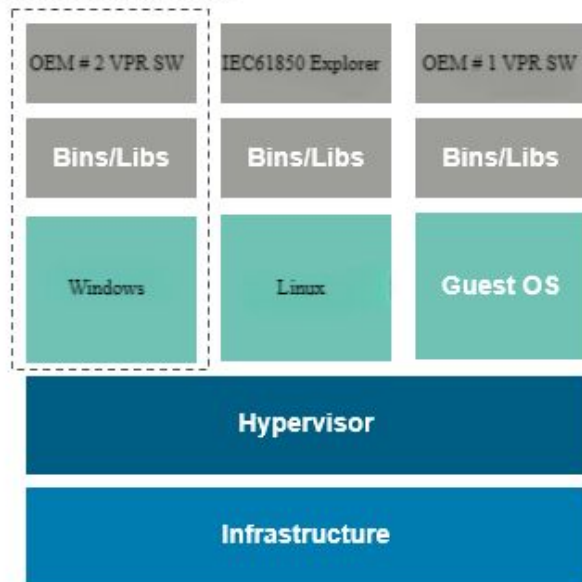
- Système d'exploitation (OS) complet
- Hyperviseur
- RAM & mémoire alloué
- Peut partager la même mémoire qu'une autre VM
- Peut être utilisé dans un autre os



Hypervisor Type 1

„bare metal hypervisor“

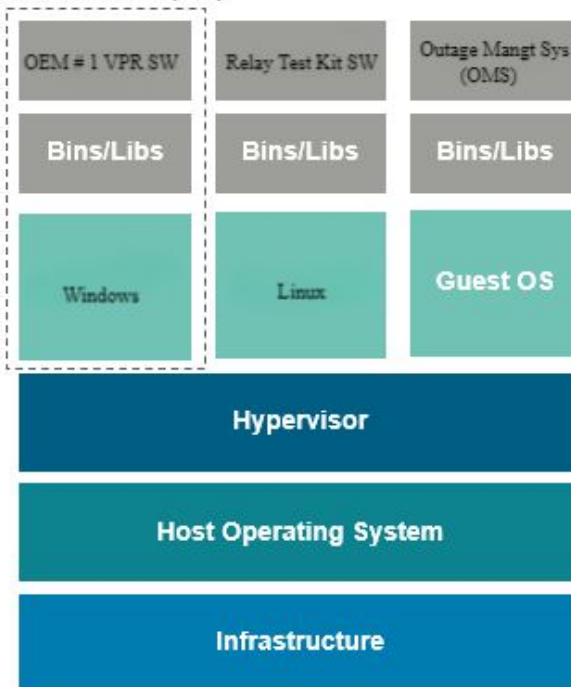
Virtual Machine (VM)



Hypervisor Type 2

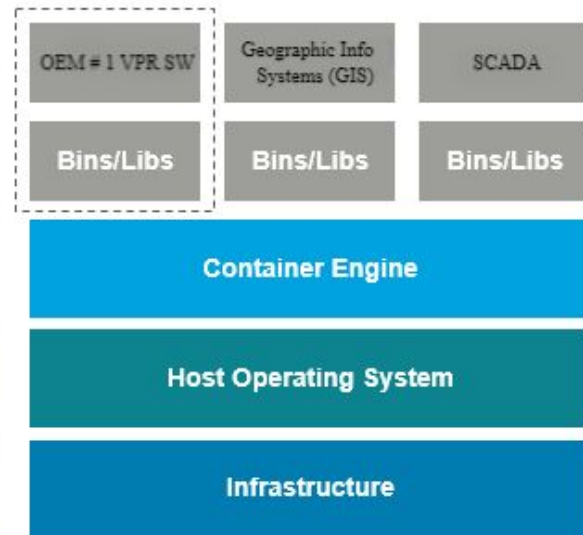
„hosted hypervisor“

Virtual Machine (VM)



Container

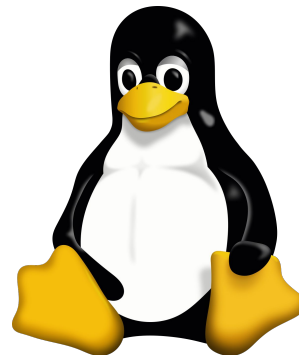
Container



Hyperviseur de type 1

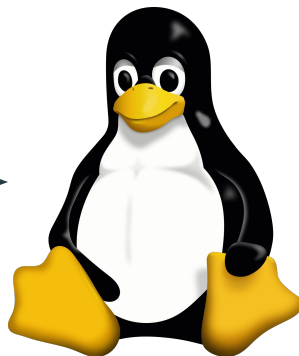


01110101



La VM me squatt

Hyperviseur de type 2

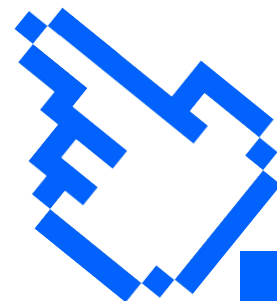


Maybe



01110101

J'peux faire des calculs stp ?



Containers :

- Une application et ses dépendances
- Execution d'une image built

Volume :

- Stockage a longue durée

App :

- Fichier source qui définissent le type de container

Bins/Lin :

- fichiers permettant d'exécuter l'application

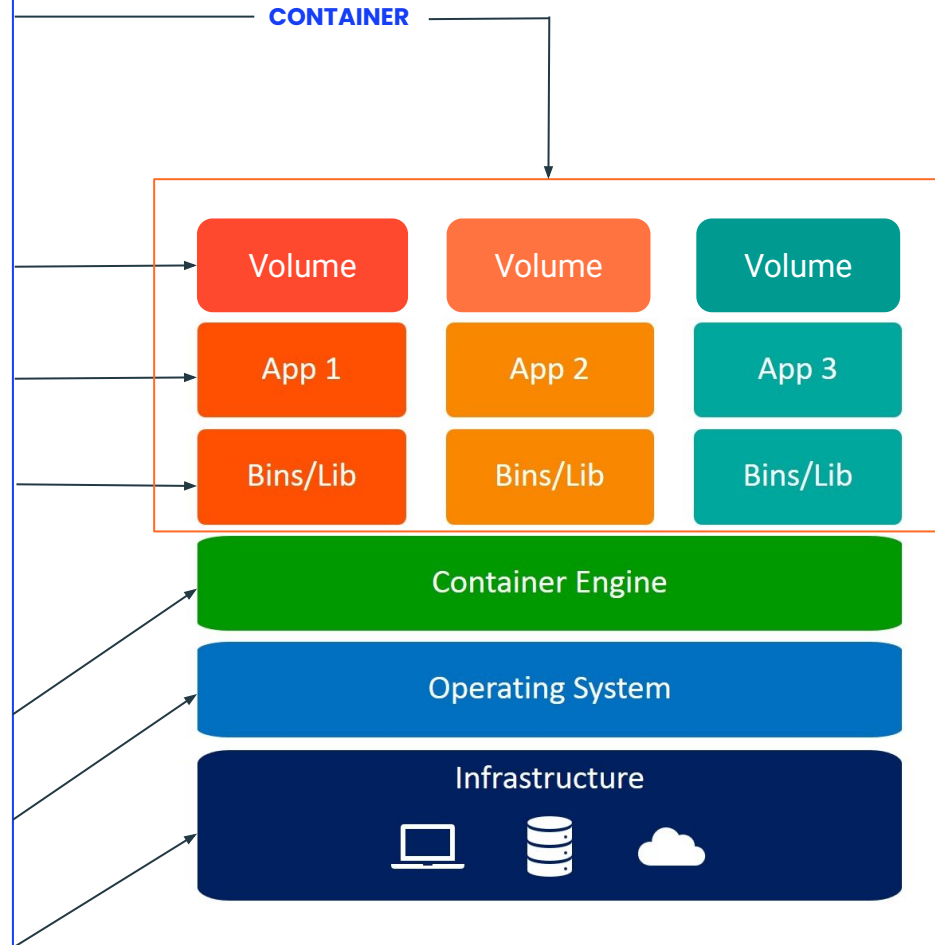
Container Engine :

- Docker Engine logiciel de gestion des container.
- C'est le composant central de Docker.

Système d'exploitation (OS)

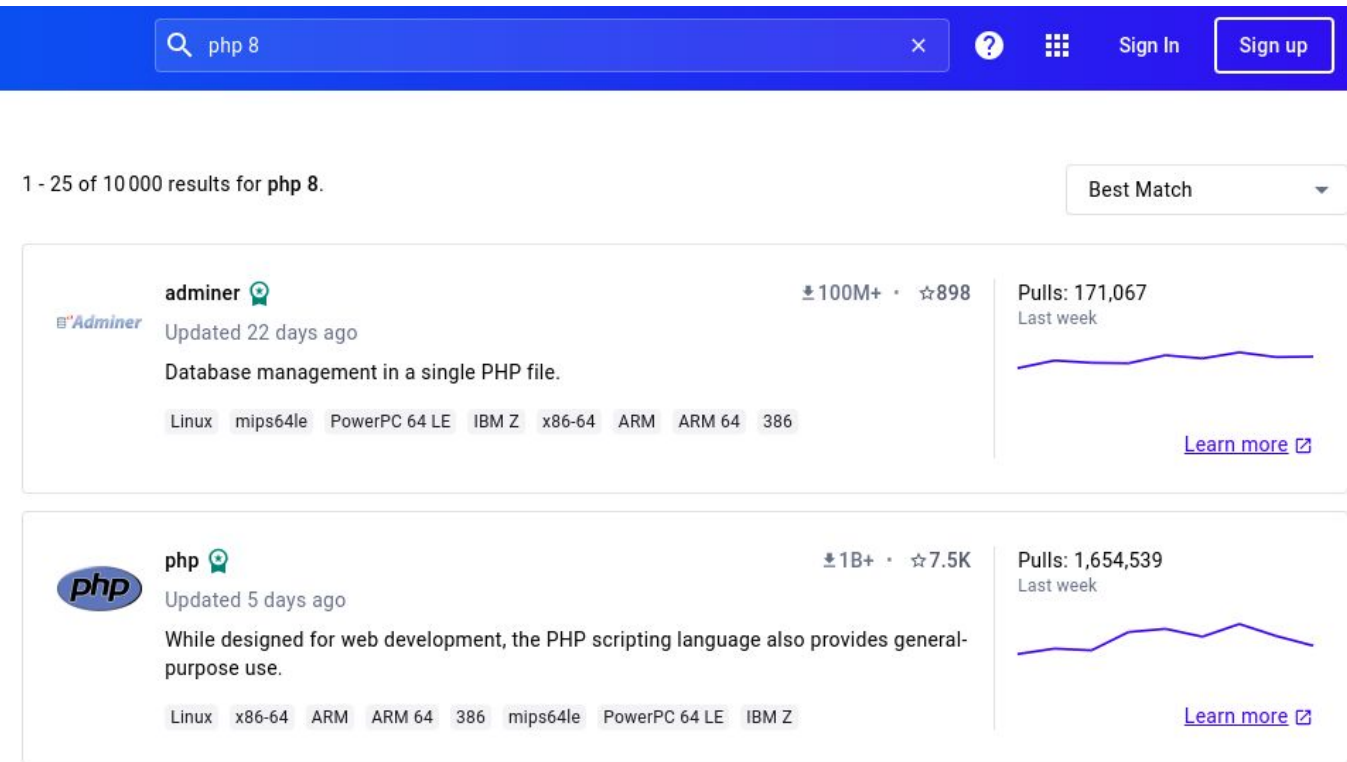
Infrastructure :

- Environnement physique (serveur, VM)



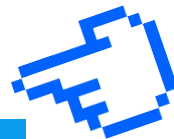
Comment Run un container ?

Utiliser une image distante



The screenshot shows the Docker Hub search results for 'php 8'. The search bar at the top contains 'php 8'. Below the search bar, it says '1 - 25 of 10000 results for php 8.' and 'Best Match' is selected. Two results are shown:

- adminer**: Updated 22 days ago. Database management in a single PHP file. Linux, mips64le, PowerPC 64 LE, IBM Z, x86-64, ARM, ARM 64, 386. Pulls: 171,067 Last week. [Learn more](#)
- php**: Updated 5 days ago. While designed for web development, the PHP scripting language also provides general-purpose use. Linux, x86-64, ARM, ARM 64, 386, mips64le, PowerPC 64 LE, IBM Z. Pulls: 1,654,539 Last week. [Learn more](#)



Analogue aux
packages

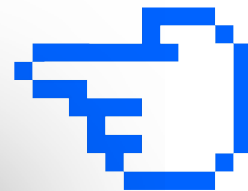
- npm
- composer

Comment Run un container ?

Utilisation d'une image distante



```
docker pull nom_utilisateur/mon_image:version  
docker run -d nom_utilisateur/mon_image:version
```



Comment run un container ?

```
# Utilise l'image PHP avec Apache de version 8.2 comme base
FROM php:8.2-apache

# Copie index.php dans le répertoire /var/www/html/ du conteneur
COPY index.php /var/www/html/

# Copie 000-default.conf dans /etc/apache2/sites-available/
COPY 000-default.conf /etc/apache2/sites-available/

# Mise à jour du système et install nano
RUN apt-get update && apt-get install -y nano

# Exposition du port 80 pour le serveur Apache
EXPOSE 80

# Commande par défaut pour démarrer le serveur Apache
CMD ["apache2-foreground"]
```

Création d'une image
locale avec Dockerfile

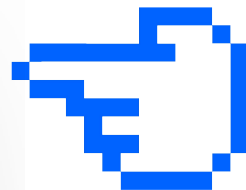
J'associe des images
distantes dans mon
image locale

Comment Run un container ?

Je build l'image locale avant de la run



```
docker build -t mon_image:latest .  
docker run -d mon_image:latest
```



Vérifier que les containers fonctionnent

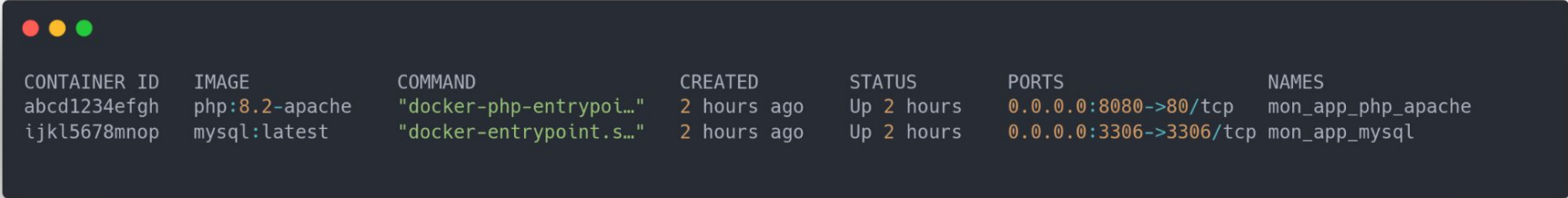
Exemple

docker ps

voir les container en train de run

docker ps -a

voir tous les container

A terminal window with a dark background and three colored window control buttons (red, yellow, green) in the top-left corner. It displays the output of the 'docker ps' command, showing two running containers. The output is organized into columns: CONTAINER ID, IMAGE, COMMAND, CREATED, STATUS, PORTS, and NAMES.

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
abcd1234efgh	php:8.2-apache	"docker-php-entrypoi..."	2 hours ago	Up 2 hours	0.0.0.0:8080->80/tcp	mon_app_php_apache
ijkl5678mnop	mysql:latest	"docker-entrypoint.s..."	2 hours ago	Up 2 hours	0.0.0.0:3306->3306/tcp	mon_app_mysql

Docker compose

Services:

db (sql)
phpmyadmin,

Définition de l'image & ses variables

port:

exposé_sur_mon_ordi:interne_à_docker

network:

permet aux container de communiquer
entre eux facilement

```
# version: '3.8'
services:
  db:
    image: mariadb:latest
    environment:
      MYSQL_DATABASE: safe_base
      MYSQL_USER: user
      MYSQL_PASSWORD: password
      MYSQL_ROOT_PASSWORD: rootpassword
    volumes:
      - db-data:/var/lib/mysql
      - ./init.sql:/docker-entrypoint-initdb.d/init.sql
    ports:
      - "3307:3306"
    networks:
      - app-network

  phpmyadmin:
    image: phpmyadmin:latest
    environment:
      PMA_HOST: db
      MYSQL_ROOT_PASSWORD: rootpassword
    ports:
      - "8081:80"
    depends_on:
      - db
    networks:
      - app-network

volumes:
  db-data:

networks:
  app-network:
```

Docker compose

volumes

```
db:
  image: mysql
  volumes:
    - db-data:/var/lib/mysql
```

```
volumes:
  - /var/lib/mysql
```

Vont être stocker ici :
`/var/lib/docker/volumes/`

```
# version: '3.8'
services:
  db:
    image: mariadb:latest
    environment:
      MYSQL_DATABASE: safe_base
      MYSQL_USER: user
      MYSQL_PASSWORD: password
      MYSQL_ROOT_PASSWORD: rootpassword
    volumes:
      - db-data:/var/lib/mysql
      - ./init.sql:/docker-entrypoint-initdb.d/init.sql
    ports:
      - "3307:3306"
    networks:
      - app-network

  phpmyadmin:
    image: phpmyadmin:latest
    environment:
      PMA_HOST: db
      MYSQL_ROOT_PASSWORD: rootpassword
    ports:
      - "8081:80"
    depends_on:
      - db
    networks:
      - app-network

volumes:
  db-data:

networks:
  app-network:
```

Récapitulatif / Questions

- VM
 - Hyperviseur de type 1
 - Hyperviseur de type 2
- Docker
 - Structure d'un container
 - Comment run un container